

Multi-variate Gaussian distributions

Random variables X and Y are jointly Gaussian if their joint PDF is given by

$$f_{XY}(x, y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho_{XY}^2}} e^{-\frac{\left(\frac{x-\mu_X}{\sigma_X}\right)^2 - 2\rho_{XY}\left(\frac{x-\mu_X}{\sigma_X}\right)\left(\frac{y-\mu_Y}{\sigma_Y}\right) + \left(\frac{y-\mu_Y}{\sigma_Y}\right)^2}{2(1-\rho_{XY}^2)}}$$

$$\begin{aligned} f_X(x) &= \frac{1}{\sqrt{2\pi\sigma_X^2}} e^{-\frac{u^2}{2}} \\ &= \frac{1}{\sqrt{2\pi\sigma_X^2}} e^{-\frac{(x-\mu_X)^2}{2\sigma_X^2}}. \end{aligned}$$

$$f_Y(y) = \frac{1}{\sqrt{2\pi\sigma_Y^2}} e^{-\frac{(y-\mu_Y)^2}{2\sigma_Y^2}}.$$

$$f_{Y|X}(y|x) = \frac{1}{\sqrt{2\pi\sigma_Y^2(1-\rho_{XY}^2)}} e^{-\frac{\left[y - \mu_Y - \rho_{XY} \frac{\sigma_Y}{\sigma_X}(x - \mu_X)\right]^2}{2\sigma_Y^2(1-\rho_{XY}^2)}},$$

where

$$\mu_{Y|X} = \mu_Y + \rho_{XY} \frac{\sigma_Y}{\sigma_X}(x - \mu_X)$$

$$\sigma_{Y|X}^2 = \sigma_Y^2(1 - \rho_{XY}^2).$$

$$\mu_{X|Y} = \mu_X + \rho_{XY} \frac{\sigma_X}{\sigma_Y}(y - \mu_Y)$$

$$\sigma_{X|Y}^2 = \sigma_X^2(1 - \rho_{XY}^2).$$

Example 7.1. At a certain university, the joint probability density function of X and Y , the grade point averages of a student in his/her freshman and senior years, respectively, is jointly Gaussian. From the grades of the past years, it is known that $\mu_X = 3$, $\mu_Y = 2.5$, $\sigma_X = 0.5$, $\sigma_Y = 0.4$ and $\rho_{XY} = 0.4$. Find the probability that a student with grade point average 3.5 in his/her freshman year will earn a grade point average of at least 3.2 in his/her senior year.

and standard deviation

$$\begin{aligned}\mu_{Y|X} &= \mu_Y + \rho_{XY} \frac{\sigma_Y}{\sigma_X} (x - \mu_X) \\ &= 2.5 + 0.4 \frac{0.4}{0.5} (3.5 - 3.0) \\ &= 2.66.\end{aligned}$$

$$\begin{aligned}\sigma_{Y|X} &= \sigma_Y \sqrt{(1 - \rho_{XY}^2)} \\ &= 0.4 \sqrt{1 - (0.4)^2} \\ &= 0.37.\end{aligned}$$

$$\begin{aligned}P[Y \geq 3.2 | X = 3.5] &= P\left[\frac{Y - 2.66}{0.37} \geq \frac{3.2 - 2.66}{0.37} | X = 3.5\right] \\ &= P[Z \geq 1.46 | X = 3.5] \\ &= 1 - \phi(1.46) \\ &\approx 1 - 0.9279 = 0.0721.\end{aligned}$$

Let us first assume that the two dimensional random vector is

$$\mathbf{X} = \begin{bmatrix} X_1 & X_2 \end{bmatrix}^T$$

with expectation vector

$$\boldsymbol{\mu}_{\mathbf{X}} = \begin{bmatrix} \mu_1 & \mu_2 \end{bmatrix}^T$$

and the covariance matrix

$$\mathbf{C}_{\mathbf{X}} = \begin{bmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{bmatrix}$$

Note that the joint Gaussian PDF of X and Y is given by

$$f_{XY}(x, y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho_{XY}^2}} e^{-\frac{\left(\frac{x-\mu_X}{\sigma_X}\right)^2 - 2\rho_{XY}\left(\frac{x-\mu_X}{\sigma_X}\right)\left(\frac{y-\mu_Y}{\sigma_Y}\right) + \left(\frac{y-\mu_Y}{\sigma_Y}\right)^2}{2(1-\rho_{XY}^2)}}$$

Now the joint Gaussian PDF can be given by

$$f_{\mathbf{X}}(\mathbf{x}) = \frac{1}{\sqrt{(2\pi)^2 |\mathbf{C}_{\mathbf{X}}|}} e^{-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu}_{\mathbf{X}})^T \mathbf{C}_{\mathbf{X}}^{-1}(\mathbf{x}-\boldsymbol{\mu}_{\mathbf{X}})}$$

The multivariate joint PDF of random vector $\mathbf{X}$ is Gaussian, if its PDF is

$$f_{\mathbf{X}}(\mathbf{x}) = \frac{1}{(2\pi)^{N/2} |\mathbf{C}_{\mathbf{X}}|^{1/2}} e^{-(\mathbf{x}-\boldsymbol{\mu}_{\mathbf{X}})^T \mathbf{C}_{\mathbf{X}}^{-1}(\mathbf{x}-\boldsymbol{\mu}_{\mathbf{X}})}$$

$$\begin{aligned}
\mathbf{C}_X &= \begin{bmatrix} \text{Var}[X_1] & \text{Cov}[X_1, X_2] & \cdots & \text{Cov}[X_1, X_N] \\ \text{Cov}[X_2, X_1] & \text{Var}[X_2] & \cdots & \text{Cov}[X_2, X_N] \\ \cdots & \cdots & \cdots & \cdots \\ \text{Cov}[X_N, X_1] & \text{Cov}[X_N, X_2] & \cdots & \text{Var}[X_N] \end{bmatrix} \\
&= \begin{bmatrix} \sigma_{X_1}^2 & 0 & \cdots & 0 \\ 0 & \sigma_{X_2}^2 & \cdots & 0 \\ \cdots & \cdots & \cdots & \cdots \\ 0 & 0 & \cdots & \sigma_{X_N}^2 \end{bmatrix}
\end{aligned}$$

The quadratic form that appears in the exponent of the Gaussian PDF is given by

$$\begin{aligned}(\mathbf{x} - \boldsymbol{\mu}_X)^T \mathbf{C}_X^{-1} (\mathbf{x} - \boldsymbol{\mu}_X) &= \begin{bmatrix} x_1 - \mu_1 & \cdots & x_N - \mu_N \end{bmatrix} \begin{bmatrix} \sigma_1^{-2} & \cdots & 0 \\ \cdots & \cdots & \cdots \\ 0 & 0 & \sigma_N^{-2} \end{bmatrix} \begin{bmatrix} x_1 - \mu_1 \\ \vdots \\ x_N - \mu_N \end{bmatrix} \\ &= \sum_{n=1}^N \left(\frac{x_n - \mu_n}{\sigma_n} \right)^2.\end{aligned}$$

Example 7.4. Consider the outdoor temperature at a certain weather station. On a certain day, the temperature measurements in units of degree Fahrenheit taken at 6 am, 12 noon, and 6 pm are all Gaussian random variables, X_1 , X_2 , and X_3 with variance 16 degree². The expected values are 50 degrees, 62 degrees and 58 degrees, respectively. The covariance matrix of the three measurements is given by

$$\mathbf{C} = \begin{bmatrix} 16.0 & 12.8 & 11.2 \\ 12.8 & 16.0 & 12.8 \\ 11.2 & 12.8 & 16.0 \end{bmatrix}$$

$$\mu_1 = 50$$

$$\mu_2 = 62$$

$$\sigma_1^2 = \sigma_2^2 = 16.0$$

$$\text{Cov}[X_1, X_2] = 12.8$$

$$\begin{aligned}\rho_{12} &= \frac{\text{Cov}[X_1, X_2]}{\sigma_1 \sigma_2} \\ &= \frac{12.8}{16} \\ &= 0.8.\end{aligned}$$

$$f_{X_1, X_2}(x_1, x_2) = \frac{\exp \left[-\frac{(x_1 - 50)^2 - 1.6(x_1 - 50)(x_2 - 62) + (x_2 - 62)^2}{19.2} \right]}{60.3}.$$

$$\boldsymbol{\mu}_{\mathbf{W}} = \begin{bmatrix} 50 \\ 62 \end{bmatrix}, \quad \mathbf{C}_{\mathbf{W}} = \begin{bmatrix} 16.0 & 12.8 \\ 12.8 & 16.0 \end{bmatrix}.$$

$$\det(\mathbf{C}_{\mathbf{W}}) = 92.16 \text{ and } \det(\mathbf{C}_{\mathbf{W}})^{1/2} = 9.6.$$

$$f_{\mathbf{W}}(\mathbf{w}) = \frac{1}{60.3} \exp \left(-\frac{1}{2} (\mathbf{w} - \boldsymbol{\mu}_{\mathbf{W}})^T \mathbf{C}_{\mathbf{W}}^{-1} (\mathbf{w} - \boldsymbol{\mu}_{\mathbf{W}}) \right).$$

Given an n -dimensional Gaussian random vector $\mathbf{X}$ with expected value $\boldsymbol{\mu}_{\mathbf{X}}$ and covariance $\mathbf{C}_{\mathbf{X}}$, and an $m \times n$ matrix $\mathbf{A}$ with $\text{rank}(\mathbf{A}) = m$,

$$\mathbf{Y} = \mathbf{A}\mathbf{X} + \mathbf{b}$$

is an m -dimensional Gaussian random vector with expected value $\boldsymbol{\mu}_{\mathbf{Y}} = \mathbf{A}\boldsymbol{\mu}_{\mathbf{X}} + \mathbf{b}$ and covariance $\mathbf{C}_{\mathbf{Y}} = \mathbf{A}\mathbf{C}_{\mathbf{X}}\mathbf{A}'$.

Example 8.11

Continuing Example 8.10, use the formula $Y_i = (5/9)(X_i - 32)$ to convert the three temperature measurements to degrees Celsius.

- (a) What is $\mu_{\mathbf{Y}}$, the expected value of random vector $\mathbf{Y}$?
- (b) What is $\mathbf{C}_{\mathbf{Y}}$, the covariance of random vector $\mathbf{Y}$?
- (c) Write the joint PDF of $\mathbf{Y} = [Y_1 \ Y_2 \ Y_3]'$ using vector notation.

(a) In terms of matrices, we observe that $\mathbf{Y} = \mathbf{A}\mathbf{X} + \mathbf{b}$ where

$$\mathbf{A} = \begin{bmatrix} 5/9 & 0 & 0 \\ 0 & 5/9 & 0 \\ 0 & 0 & 5/9 \end{bmatrix}, \quad \mathbf{b} = -\frac{160}{9} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}.$$

(b) Since $\mu_X = [50 \ 62 \ 58]'$, from Theorem 8.11,

$$\mu_Y = A\mu_X + b = \begin{bmatrix} 10 \\ 50/3 \\ 130/9 \end{bmatrix}.$$

(c) The covariance of $\mathbf{Y}$ is $\mathbf{C}_Y = \mathbf{A}\mathbf{C}_X\mathbf{A}'$. We note that $\mathbf{A} = \mathbf{A}' = (5/9)\mathbf{I}$ where $\mathbf{I}$ is the 3×3 identity matrix. Thus $\mathbf{C}_Y = (5/9)^2\mathbf{C}_X$ and $\mathbf{C}_Y^{-1} = (9/5)^2\mathbf{C}_X^{-1}$. The PDF of $\mathbf{Y}$ is

$$f_Y(\mathbf{y}) = \frac{1}{24.47} \exp \left(-\frac{81}{50} (\mathbf{y} - \boldsymbol{\mu}_Y)^T \mathbf{C}_X^{-1} (\mathbf{y} - \boldsymbol{\mu}_Y) \right). \quad (8.57)$$

